Machine Learning to Reduce Food Wastage in Supermarkets

A Data Science Project Using NumPy, Pandas & Machine Learning to Support Sustainable Food Systems

B. Poojitha
192424374

1. Problem Understanding

Problem Statement

Supermarkets frequently overstock perishable items such as fruits, vegetables, dairy products, and bakery goods. Because demand is difficult to predict accurately, unsold stock expires and is discarded. This results in food waste, direct financial loss for retailers, and unnecessary environmental impact from producing food that is never consumed.

Objectives

- Predict daily demand for perishable products using Machine Learning.
- Reduce food wastage by optimizing inventory levels.
- Improve product availability while minimizing overstock.
- Support United Nations Sustainable Development Goal 2 – Zero Hunger by reducing food waste and improving food distribution.

2. Data Collection & Preparation

Dataset Features

Feature	Description
Date	Sale date
Product_ID	Product code
Product_Name	Item name
Category	Fruits, Dairy, Bakery, etc.
Stock	Available stock
Sales	Units sold
Price	Selling price
Discount	Discount percentage
Temperature	Weather temperature
Holiday	Holiday indicator
Expiry_Days	Days before expiry
Waste	Unsold quantity

Data Preprocessing

- Remove missing values.
- Remove duplicate records.
- Convert date into month/day.
- Encode categorical variables.
- Normalize numerical values.
- Split dataset into training and testing sets.

3. Analysis using NumPy and Pandas

Import Libraries

```
import pandas as pd
import numpy as np
```

Load Dataset

```
df = pd.read_csv("supermarket_food.csv")
```

Display Dataset

```
df.head()
```

Check Missing Values

```
df.isnull().sum()
```

Summary Statistics

```
df.describe()
```

Total Food Waste

```
df["Waste"].sum()
```

Average Sales

```
df["Sales"].mean()
```

Category-wise Waste

```
df.groupby("Category")["Waste"].sum()
```

Machine Learning Model

Example using Random Forest Regression to predict daily sales from stock, price, discount and temperature:

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor

X = df[['Stock','Price','Discount','Temperature']]
y = df['Sales']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

model = RandomForestRegressor()
model.fit(X_train, y_train)

prediction = model.predict(X_test)
```

4. Visualization & Interpretation

1. Food Waste by Category

[CHART] Bar Chart

Total waste quantity summed per product category.

Insight: Shows which product category wastes the most food.

2. Monthly Sales Trend

[CHART] Line Graph

Total or average sales plotted across months.

Insight: Identifies seasonal demand patterns.

3. Waste vs Stock

[CHART] Scatter Plot

Waste quantity plotted against stock levels for each record.

Insight: Higher stock generally results in higher waste.

4. Correlation Heatmap

[CHART] Heatmap

Correlation matrix across Sales, Stock, Discount, Temperature, and Waste.

Shows relationships among:

- Sales
- Stock
- Discount
- Temperature
- Waste

5. Actual vs Predicted Sales

[CHART] Line Graph

Overlay of actual test-set sales values against the Random Forest model's predictions.

Shows prediction accuracy of the ML model.

5. Innovation & SDG Mapping

Innovation

The proposed system uses Machine Learning to:

- Predict future product demand.
- Recommend optimal stock levels.
- Alert staff about products nearing expiry.
- Suggest discounts for slow-moving products.
- Recommend food donation before

expiration. This reduces both food waste and financial losses.

SDG Mapping — SDG 2: Zero Hunger

The project supports SDG 2 by:

- Reducing edible food wastage.
- Improving food availability.
- Encouraging redistribution of surplus food.
- Increasing efficiency in food supply chains.
- Supporting sustainable consumption practices.

6. Documentation & Conclusion

Conclusion

Machine Learning enables supermarkets to accurately forecast customer demand and optimize inventory management. By reducing overstocking and identifying products nearing expiry, supermarkets can significantly decrease food wastage, lower operational costs, and improve customer satisfaction. The system also supports SDG 2 – Zero Hunger by promoting efficient food distribution and reducing the loss of edible food.

Expected Output

- Accurate demand prediction.
- Reduced expired products.
- Lower inventory costs.
- Improved supermarket efficiency.
- Less food wastage.
- Contribution to SDG 2 – Zero Hunger.